The LaTeX report

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1 Setup

1.1 Command history

```
ma5>import /data/ammelsayed/phase_space_biasing/runs/z012j_MLM/bin/internal/ufomodel
ma5>import /data/ammelsayed/phase_space_biasing/runs/z012j_MLM/Events/run_01/unweighted_events.lhe
as run_01
ma5>define vl = 12 14 16
ma5>define vl  = -16 -14 -12
ma5>define nu = -16 -14 -12 12 14 16
ma5>define invisible = vt vm  vt  ve  ve vm vl vl  nu
ma5>set main.graphic_render = matplotlib
ma5>plot THT 40 0 500 [logY]
ma5>plot MET 40 0 500 [logY]
ma5>plot SQRTS 40 0 500 [logY]
ma5>plot PT(l+[1]) 40 0 500 [logY]
ma5>plot ETA(l+[1]) 40 -10 10 [logY]
ma5>plot PT(l-[1]) 40 0 500 [logY]
ma5>plot ETA(l-[1]) 40 -10 10 [logY]
ma5>plot M(l+[1] l-[1]) 40 0 500 [logY ]
ma5>plot DELTAR(l+[1],l-[1]) 40 0 10 [logY ]
ma5>submit /data/ammelsayed/phase_space_biasing/runs/z012j_MLM/MA5_PARTON_ANALYSIS_analysis1
```

1.2 Configuration

- MadAnalysis version 1.11.0 (2026-07-26).
- Histograms given for an integrated luminosity of 10fb^{-1}.

2 Datasets

2.1 run_01

- Sample consisting of: signal events.
- Generated events: 100000 events.
- Normalization to the luminosity: 8120+/- 7 events.
- Ratio (event weight): 0.081 .

Path to the event file	Nr. of events	Cross section (pb)	Negative wgts (%)
z012j_MLM/Events/run_01/-unweighted_events.lhe.gz	100000	0.812 @ 0.086%	0.0

3 Histos and cuts

3.1 Histogram 1

*** Plot: THT**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	8120	1.0	0.0	0.0	0.0	0.0

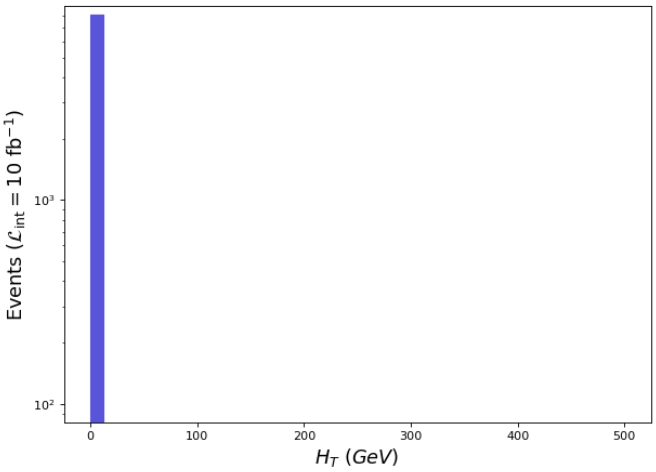


Figure 1.

3.2 Histogram 2

*** Plot: MET**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	8120	1.0	0.0	0.0	0.0	0.0

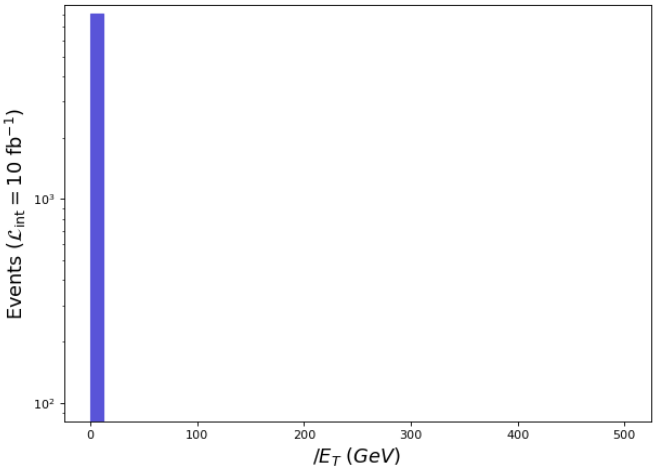


Figure 2.

3.3 Histogram 3

*** Plot: SQRTS**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	8120	1.0	470.782	185.2	0.0	28.4

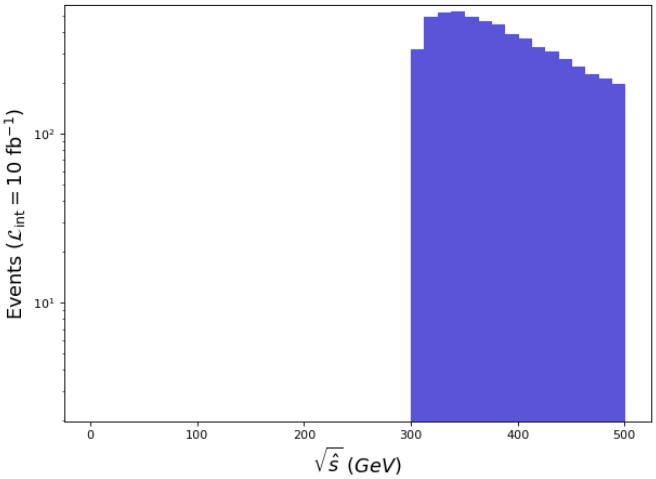


Figure 3.

3.4 Histogram 4

*** Plot: PT (l+[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	209.675	75.08	0.0	1.076

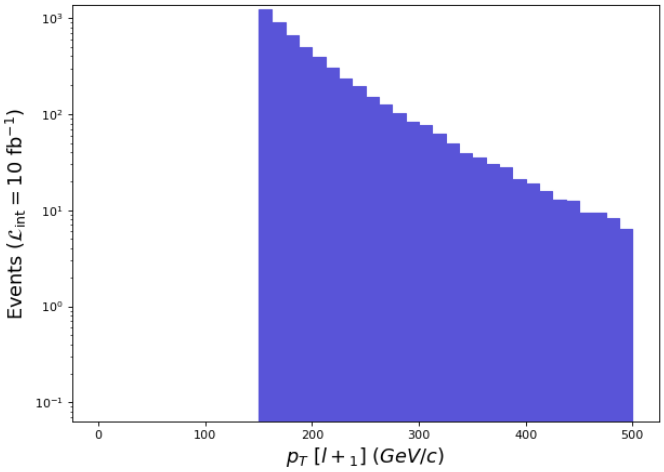


Figure 4.

3.5 Histogram 5

*** Plot: ETA (l+[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	0.00157704	1.179	0.0	0.0

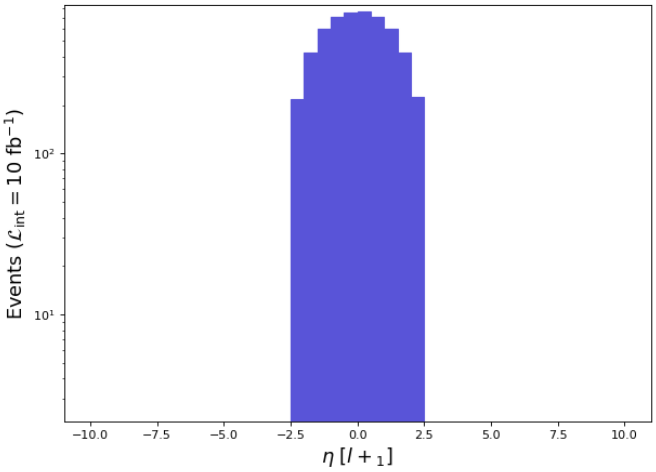


Figure 5.

3.6 Histogram 6

*** Plot: PT (l-[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	209.675	75.08	0.0	1.076

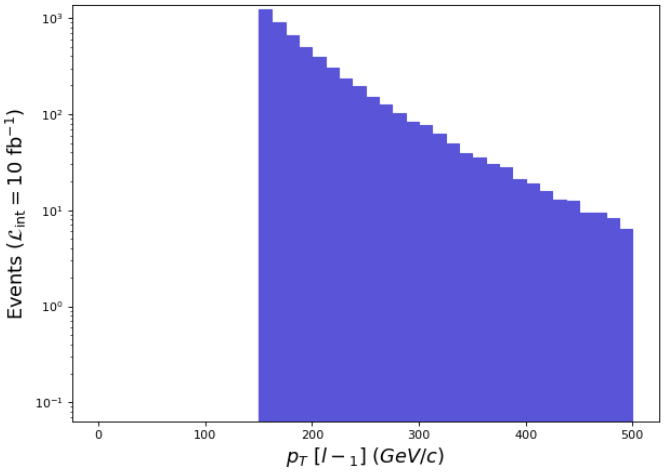


Figure 6.

3.7 Histogram 7

*** Plot: ETA (l-[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	-0.0012902	1.377	0.0	0.0

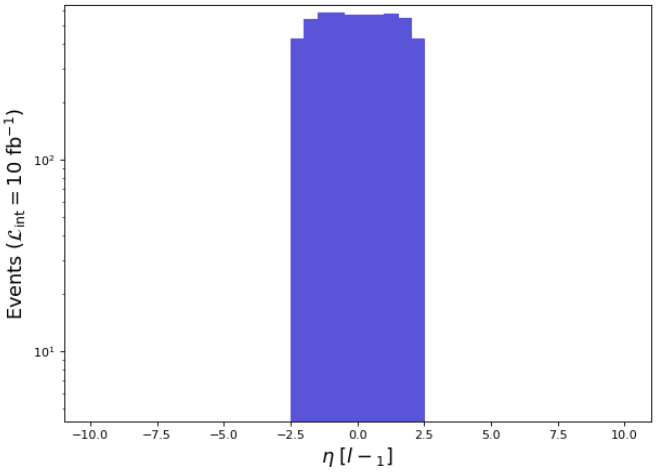


Figure 7.

3.8 Histogram 8

*** Plot: M (l+[1] l-[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	471.073	186.0	0.0	28.36

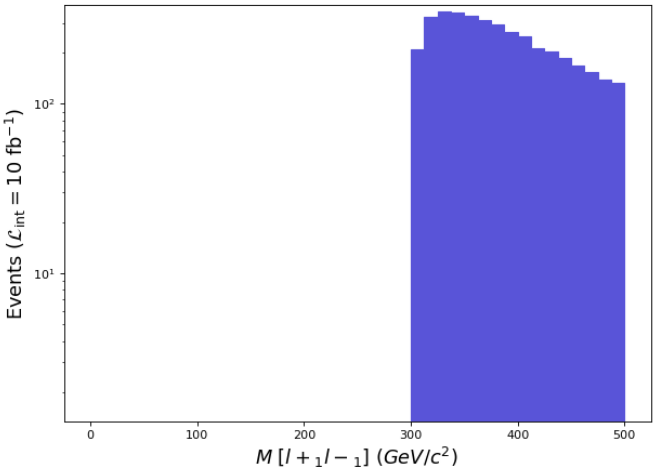


Figure 8.

3.9 Histogram 9

*** Plot: DELTAR (l+[1] , l-[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	5411	1.0	3.28139	0.1929	0.0	0.0

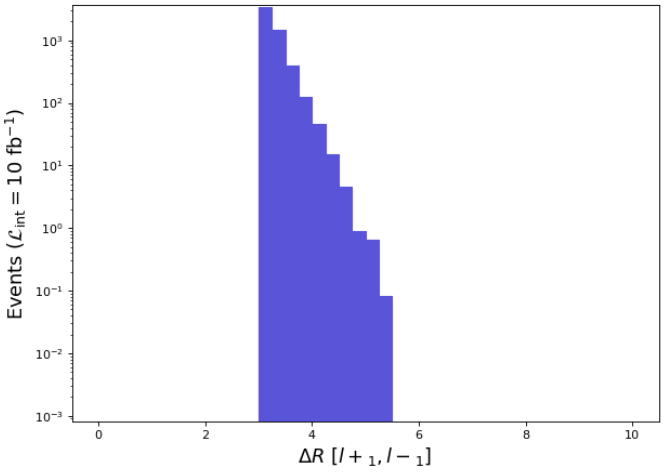


Figure 9.